Safety : It is state or condition to be safe from illness (family) injury or loss. It is used to denote risk.

(i) Identify the risk

(ii) Potential Cause

(iii) Eliminate those causes.

Risk : Safety inversely proportional to Risk

$$\text{Safety} \propto \frac{1}{\text{Risk}}$$

→ Risk May be defined as the Hazard associated with event process or system.

Hazard :- where Hazard is the potential intent to cause harm to the human body in the form of illness, injury or totality, or to the system by changing its state.

Safety
- social safety (Behaviour among people psychology)
- physical safety (Prob. of being killed / injured loss)

Accident :- An unwanted, unplanned & unexpected event. Occuring through a sequence of events. leading to an undesirable consequences.

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Incident :- Danger occurance

→ May or may not lead to an undesirable consequences.

Injury :- → Fatal (death)

→ Non fatal → Permanent (

→ Temp. disability

Serious Injury :- Permanent loss of a body part

Reportable injury :- Report under 72 hours

Minor injury :- 24 hours to 72 hrs return to work

Disaster : More than 10 fatal injuries in an accident thus we say disaster

→ In a particular accident 10 or more person are died then it is known as disaster

---- → 4-9 Major accident

Mi.. → fatal injury ~~More~~ less than 1 fatal. accident.

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Evaluation of Safety performance :-

(1) Injury rate per 1000 person employed

$$= \frac{\text{No. of injuries in a given year}}{\text{Avg daily employment}} \times 1000$$

- fatality is also a injury untill and unless it is specified

(2) Injury rate per Million tonne output

$$= \frac{\text{No. of injuries in a given time period}}{\text{total production (in te.) during that period}} \times 10^6$$

(3) Injury per one lakh Man shifts worked

$$= \frac{\text{No. of injuries in a given time period}}{\text{total Man-shift worked during that period}} \times 10^5$$

(4) Injury rate per million Man-hours worked

$$= \frac{\text{No. of injuries in a given time period}}{\text{total Man-hours worked during that period}} \times 10^6$$

(5) Accident Severity Rate (ASR)

$$= \frac{\text{Total time charges (in days)}}{\text{total Man-hours worked}} \times 10^6$$

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= Time charged [for fatalities · serious bodily injury + ~~resp~~ · reportable injury + Minor injury (days)]

Total Man hours Worked

for fatalities → time charges = 6000 days

permanent total disability = 6000 days
= 4500 days

· Severity Index (IFRM) = $\frac{50F + S}{\text{Total Man-Shifts}} \times 10^5$

where, F = Total no. of fatalities

S = Total no. of serious bodily injuries

1 fatal injury = 50 serious bodily injury.

LAFP = (Largest Accident free period)

IFRM = (Injury frequency rate of Manshift)

(i) frequency rates are reactive, not predictive

(ii) Accidents are rare events and so frequency values on them may not be statistically valid

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$\times 10^6$

(iii) Frequency rates can be manipulated

(iv) Too much focus on severity rates / indices false security.

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How Can you prepare a representative index.

prepare a Model which will provide a index.

DOMino Theory / Heinrich Philosophy

1931 : H.W. Heirich → Engg & Inspection division of traveller's insurance Company published a book Industrial accident preventation.

88 % Total accident = Unsafe acts of people (human behavior)

10 % total accident = Unsafe Condition

2 % total accidents = Act of God

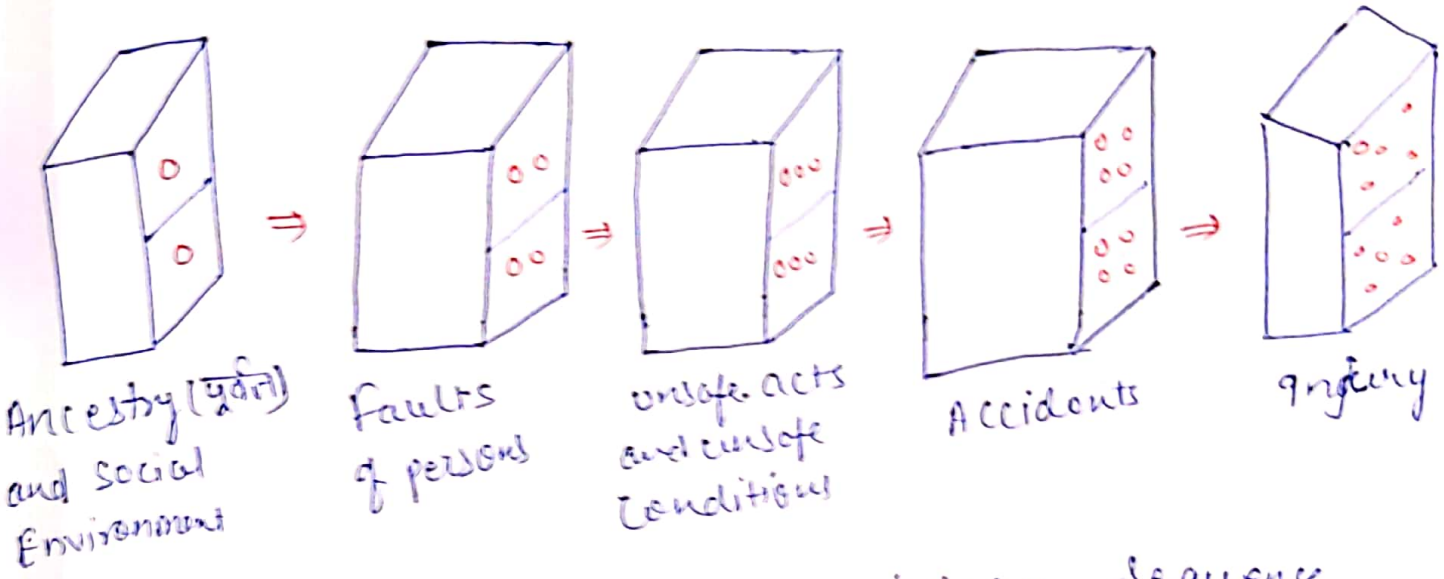
Heinrich's philosophy for Causation of Accidents Domino theory.

→ Prior (gf) to 1931, it was believed that accidents are Caused by unsafe Condition.

But in 1931 H.W Heinrich Published a book on accidents are Mostly being Caused by the unsafe acts of the people rather than the unsafe conditions.

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After the people have emphasized on ill-human behaviour at workplace & to control the industrial accidents.



five factors in the injury sequence

Examples of unsafe acts

1. Operating the machine at unsafe speed
2. Making safety devices inoperative
3. Not using safety devices
4. Horseplay (Teasing distracting)
5. Storing machine without working
6. Using unsafe equipment
7. Crossing a fence
8. maintenance work on moving machines.

Examples of unsafe condition

1. PPE not available / not in operation.

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Risk Management

Management :- It take place when some thing available is limited.

Risk Management :- Risk Management is defined as the system & procedure for reducing and eliminating risk including human, organisational, environmental financial & technological risk.

- → Financial Risk
- → Safety risk
- → Environmental risk

Risk Analysis :-

- → System def^{n}
- → Identification Hazards
- → Risk Estimation.

Duality Risk :-

$$\text{Risk Rating} = \left(\text{Prob. of occurances of an event}\right) \times \left(\text{Consequences}\right)$$

$$\text{Risk Value} = \left(\text{Prob of occurance of an event}\right) \times \left(\text{Consequences}\right) \times \left(\text{Exposure}\right)$$

Triplet def^{n} of Risk) ⇒ Exposure comes in Picture when environmental risk is associated

Where,

Risk Rating (RR)

Risk value. (Rv)

a = Threshold value

$\therefore RR \geq a$

Risk Evaluation :-

→ Risk tolerability decision $RR \geq a$

→ Evaluation of alternative options.

Risk analysis.
- → System defin
- → Identification of Hazard
- → Risk Estimation

Risk assesment

\+ . = **Risk Management**

Risk Evaluation
- → Risk tolerability decision $RR \geq a$?
- → Evaluation of alternative options.

Risk Control
- → Decision Making
- → Implementation
- → Monitoring

→ Risk analysis } Risk assesment
Risk Evaluation } + Risk Control } = Risk Management

P

Risk Management process

Risk Analysis → System definition
→ identification
→ Risk Estimation

Risk Evaluation :--> Risk tolerability
→ decisions
→ Analysis of option

Risk Reduction :-
→ decision Making
→ implementation
→ Monitoring

Risk Analysis :-

- Risk Analysis is a technique for identifying characterizing Quantifying and Evaluating hazards
- It consist of two distinct phases

(1) A Qualitative step of identifying characterizing and ranking hazards

(2) A Quantitative step of risk Evaluation. which includes Estimating the likehood and consequences of hazard occurance.

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Quantitative risk analysis :-

Quantitative risk analysis (QRA) is the art and science of developing and understanding numerical estimates of the risk (i.e. combinations of the expected frequency probability / likelihood and the consequences) associated with a facility or operation.

It uses a set of highly sophisticated, but approximate tools for acquiring and understanding the risk

The QRA method can be used throughout all phases of the life of a process or a system. The primary goal of QRA is to provide just enough analysis to the risk based Question.

Qualitative risk analysis :-

Qualitative risk analysis is a technique with discovering the probability of occurance of an event and its impact if it occurs.

In Qualitative risk analysis, the method used for analysing risk components such as likelihood and its impact are subjectively evaluated because of observe sufficient

historical data or Quantitative risk analysis is time consuming or costly.

Thus assigning probability and impacts to risks becomes a subjective Exercise to develop immediate risk responses.

Evaluating the risk : Qualitative

Likelihood

① Highly unlikely
② possibly
③ Quite likely
④ very likely

Severity

① slight harm
② injury
③ Serious injury
④ possible fatality

Risk = likelihood x severity

Risk Matrix

Likelihood \ Severity	1	2	3	4
4	4 (3)	8 (2)	12 (1)	16 (1)
3	3 (4)	6 (3)	9 (2)	12 (1)
2	2 (4)	4 (3)	6 (3)	8 (2)
1	1 (4)	2 (4)	3 (4)	4 (3)

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Controlling the risk

① unacceptable stop doing it untill improvements are made. (≥ 9).

② Significant proced with caution, but improvement high priority. (7, 8, 9)

③ Tolerable okey to proced but plan to improve (4, 5, 6).

④ Insignificant Any possible improvement, low prority. (1, 2, 3)

1, 2, 3 → ~~neglect~~ avoidable.

4, 5, 6 → Tolerable.

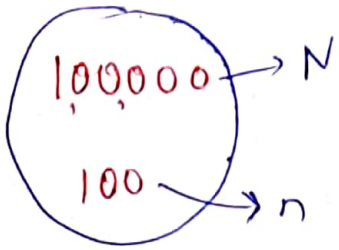
7, 8, 9 → highly prioity.

≥ 9 → unacceptable.

Sample Mean

$\bar{x} = \frac{\text{Sum}}{n}$ → Sample population

$\bar{x} \simeq \mu$



∴ If $n \uparrow$ $\bar{x} \to \mu$

Population Mean

$N = 100,000$

$\mu = \frac{\text{Sum}}{N}$

μ → Parameter

N → entire Population

n = Sample Size

Statistic :-

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{x_1 + x_2 + x_3 + x_4 \cdots x_n}{n}$$

$$\bar{H} = \frac{\sum_{i=1}^{N} x_i}{N} = \frac{x_1 + x_2 + x_3 \cdots x_N}{N}$$

Event (Set of value generated through an experiment)

- Simple event
  (Only listed and possible outcome Can occur)
- Compound event
  (occurance of More than One outcome)

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Sturge's rule :-

$$k = [1 + 3.3 \log_{10} N]$$

where, N = no. of data
k = no. of classes
(Refer to greatest integer)

$$\left\{ \text{Interval} = \frac{\text{Max} - \text{Mini}}{k} \right\}$$

Note: (1) Barchart - discreate data
(2) Histogram - Continuous (Continuation of data is given)

Years	Frequency	Prob	Cumulative freq	Cumulative Prob
30-40	5	$5/50 = 0.1$	5	$5/50 = 0.1$
40-50	10	$10/50 = 1/5$	15	$15/50 = 0.3$
50-60	20	$20/50 = 2/5$	35	$35/50 = 0.7$
60-70	10	$10/50 = 1/5$	45	$45/50 = 0.9$
70-80	5	$5/50 = 1/10$	50	$50/50 = 1$

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Cumulative probability :-

CDF of a Cumulative Distribution fn (CDF)

$F(45 \leq x \leq 48)$

$F(x \leq 48 - F(x \leq 45)$

$$\int_{-\infty}^{48} f(x)\,dx - \int_{-\infty}^{45} f(x)\,dx$$

We use PDF & CDF when data is continuous
when data is discrete.

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(1) Probability Distribution function
(2) Probabilite. density function
(3) Probability Mass function
(4) Cumulative. Distribution function



Normal



log normal

Probability distribution function

PDF (Probability density function)

Ex:- continuous Random Variable.

PMF (Probability Mass function)

Ex:- discrete. random Variable. (whole no.)

Random variable :-

→ Random variable is a process of Mapping the output of a Random process or experiment to a number. (Random variable is a variable which can be or more than one value)

Ex:- Tossing a coin [Experimently Outcome]

$$X = \begin{cases} 0 & \text{if head} \\ 1 & \text{if tail} \end{cases}$$

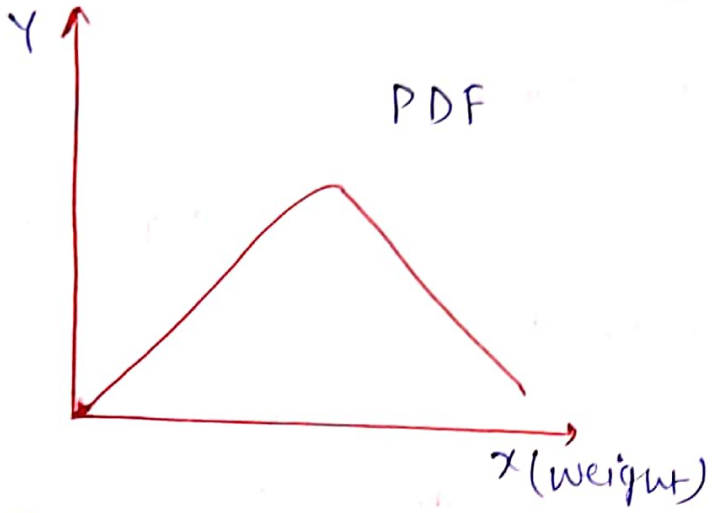
Ex:- Rolling a dice {Experiment} - {0,1,2,3,4,5,6}

y = {Sum of the outcome of rolling a dice 5 time.}

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① Probability density function (PDF) :-

① Distribution of continuous Random variable



Slope = $\frac{dx}{dy}$

$\frac{\text{Change in } x}{\text{Change in } y}$

Probability density = gradient decent of Cumulative Curve.

Slope = Gradient = Change = $\frac{dx}{dy}$

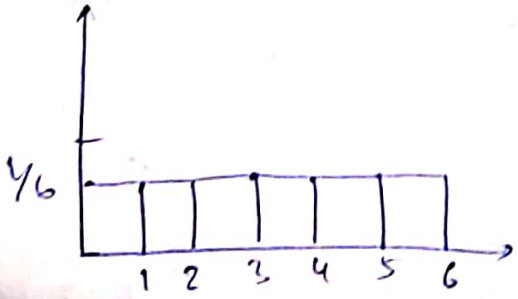
② Probability Mass function :-

Discrete Random variable

Ex:- Rolling ~~Random variable~~ a dice

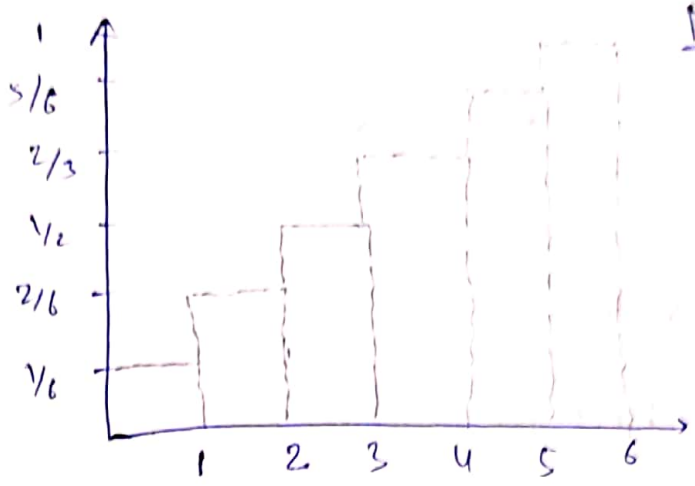
[1 2 3 ... 6] $P(1) = \frac{1}{6}$

$P(6) = 1/6$



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Cumulative density function



Now

If $P(X \leq 2) = P(X=1) + P(X=2)$

$= \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$

If $P(X \leq 6) = P(X=1) + P(X=2) + P(X=3) + P(X=4) \cdots P(X=6)$

$= \left(\frac{1}{6} + \frac{1}{6} \cdots \frac{1}{6}\right) = \frac{1}{6} \times 6$

$= 1$

PDF

For a discrete distribution

① $0 \leq P(x) \leq 1$

② $\sum_{i=1}^{n} P(x_i) = 1$

③ $\mu = \sum x_i P(x_i) = E(x)$

④ $\sigma = \sqrt{\sum_{i=1}^{n} (x_i - \mu)^2 P(x_i)}$

⑤ $F(x) = \frac{d F(x)}{dx}$

⑥ $\int_{-\infty}^{\infty} f(x)\, dx = 1$

⑦ $P(a \leq x < b) = \int_{a}^{b} f(x)\, dx$

$= F(b) - F(a)$

for Continuous distribution :-

① $0 \leq f(x) \leq 1$

② $F(x) = P(X \leq x) = \int_{-\infty}^{x} f(x)\, dx$

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(iii) $E(x) = \mu = \int_{-\infty}^{\infty} x \cdot f(x)\, dx$

(iv) $\sigma = \sqrt{\int_{-\infty}^{\infty} (x-\mu)^2 \cdot f(x)\, dx}$

(#) PMF (Discrete)

Binomial Distribution

$$P(x) = {}^{n}C_{x} \cdot p^{x} \cdot q^{(n-x)}$$

$P(x, n, p, q)$

$$E(X) = \mu = \sum_{i=1}^{n} x\, P(x)$$

$$= \sum_{i=1}^{n} (x)\, {}^{n}C_{x}\, p^{x}\, q^{(n-x)}$$

$$= \sum_{i=1}^{n} x\, \frac{n!}{x!\,(n-x)!}\, p^{x} \cdot q^{(n-x)}$$

$$= \sum_{i=1}^{n} x\, \frac{n(n-1)!}{(x-1)!\,(n-x)!}\, p \cdot p^{x-1} \cdot q^{(n-1)-(x-1)}$$

$$= \sum_{i=1}^{n} x \cdot \frac{n(n-1)!}{(n-1-x-1)!\; x(x-1)!} \cdot p \cdot p^{x-1} \cdot q^{n-1-x-1}$$

$$= np \sum_{n=1}^{n-1} \frac{(n-1)!}{\varepsilon\,(n-1-x-1)!\,(x-1)!} \cdot p^{x-1}\, q^{n-1-x-1}$$

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$= np \sum {}^{n-1}C_{n-1}\, p^{n-1} q^{n-1-n+1}$

$= np \times 1$

$= np$

$np \sum_{y=0}^{n} n_{C_y}\, p^y q^{n-y}$

$np \cdot \sum_{y=0}^{n} p(y)$

$np(1)$

$E(x) = np$

Variance = npq

(11) Underground Mining Accident Roof fall

Case description:- A Miner is working in an underground coal mine while drilling a hole for blasting the roof above him suddenly collapses trapping the miner under debrises.

Unsafe Act :

(1) failure to conduct proper roof inspection.

→ The Miner or supervisor may not have thoroughly inspected the roof before starting work.

(2) Ignoring Signs of roof instability.

→ If the Miner notice crack or sign of roof instability but continued working without reporting or addressing the issue, this is an unsafe act.

Unsafe Condition

(1) weak geological formation

→ The area where the miner was working may have had inherently weak rock formations. increasing the risk of a roof fall.

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② Inadequate roof Support :-

The Miner roof support system (bolt, prop) may have been insufficient or improperly installed failing to prevent the collapse.

② Opencast Mining Accident Vehicle Collision :-

Case Description :- In an open pit mine, a large haul truck is reversing to dump the 'OB' Material at the dumping site At the same time a smaller utility vehicle is approaching the are, due to the blindspot the truck operator doesn't see the utility vehicle, leading to a collision.

Unsafe Act :-

① Failure to follow (SOP) Standard Operating procedure.

The truck operator may have neglected to follow SOPs, such as using spotters or reversing alarm, while maneuvering vehicle.

② Inadequate communication :-

The Operators of both vehicles may not have communicated their intension leading to the collision.

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